

Recall from last class:

- A real **vector space** is a non-empty set V on which two operations, scalar multiplication and addition, are defined subject to the following 10 axioms. We call elements in V vectors. Let $u, v, w \in V$ and $c, d \in \mathbb{R}$.

1. $u + v \in V$.
2. $u + v = v + u$.
3. $(u + v) + w = u + (v + w)$.
4. there exists a vector 0 in V such that $u + 0 = u$.
5. there exists a vector $(-u)$ in V such that $u + (-u) = 0$.
6. $cu \in V$.
7. $c(u + v) = cu + cv$.
8. $(c + d)u = cu + du$.
9. $c(du) = (cd)u$.
10. $1 \cdot u = u$.

- A **subspace** of a vector space V is a subset of V such that

1. $0_V \in H$.
2. H is closed under vector addition; i.e. if $u, v \in H$, then $u + v \in H$.
3. H is closed under scalar multiplication; i.e. if $c \in \mathbb{R}$ and $u \in H$, then $cu \in H$.

- **Theorem 4.1** If $v_1, \dots, v_p \in V$, then $\text{Span}\{v_1, \dots, v_p\}$ is a subspace of V .
- We call $\text{Span}\{v_1, \dots, v_p\}$ the subspace generated by or spanned by $v_1, \dots, v_p$.
- The **null space** of an $m \times n$ matrix A is

$$\text{Null}(A) = N(A) = \{x \in \mathbb{R}^n : A\vec{x} = \vec{0}\}.$$

- **Theorem:** If A is an $m \times n$ matrix, then $N(A)$ is a subspace of $\mathbb{R}^n$.
- The **column space** of an $m \times n$ matrix A is the span of the columns of A . We denote the column space of A by $\text{Col}(A)$.
- Let V and W be vector spaces, then $T : V \rightarrow W$ is a **linear transformation** if
 1. $T(u + v) = T(u) + T(v)$ for all $u, v \in V$ and
 2. $T(cu) = cT(u)$ for all $c \in \mathbb{R}$ and $u \in V$.
- The **range or image** of T is defined by

$$\text{Range}(T) = \text{im}(T) = \{T(x) : x \in V\} = \{w \in W : \text{there exists } a \in V \text{ such that } T(a) = w\}.$$

The **kernel** of T is defined by $\ker(T) = \{x \in V : T(x) = 0_W\}$.

Definition 1. The subset $\{v_1, v_2, \dots, v_p\}$ of the vector space V is **linearly independent** if

$$c_1v_1 + \cdots + c_pv_p = 0_V$$

has only the trivial solution.

Theorem: Given $\{v_1, \dots, v_p\}$ with $p \geq 2$ and $v_1 \neq 0_V$. The vectors $v_1, \dots, v_p$ are linearly dependent if and only if for some $j > 1$, the vector v_j is a linear combination of $v_1, \dots, v_{j-1}$.

1.

- (a) Let $C[0, 2\pi]$ be the vector space of continuous real-valued functions on the interval $[0, 2\pi]$. Is $\{\sin t, \cos t\}$ linearly independent?

- (b) Consider $1 + t^2 \in \mathbb{P}_2$ and $1 - t^2 \in \mathbb{P}_2$. Are they linearly independent?

Definition 2. Let H be a subspace of a vector space V . The set $\{v_1, \dots, v_p\}$ is a **basis** for H if

- (a) $\{v_1, \dots, v_p\}$ is linearly independent,
- (b) $H = \text{Span}\{v_1, \dots, v_p\}$.

Spanning Set Theorem: Let $S = \{v_1, \dots, v_p\} \subseteq V$ and $H = \text{Span}\{v_1, \dots, v_p\}$.

- (a) Suppose v_k can be written as a linear combination of the other vectors in S , then $S - \{v_k\}$ still spans H .
- (b) If $H \neq \{0_V\}$, some subset of S is a basis for H .

2. Prove the theorem.

Theorem: The pivot columns of an $m \times n$ matrix A form a basis for $\text{Col}(A)$.

3. Prove the theorem.

4. Let $B = \begin{bmatrix} 1 & 0 & 6 & 5 \\ 0 & 2 & 5 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. Find a basis for $\text{Col}(B)$.

5. Is $\{1 + t, 1 - t, 2\}$ a basis for $\mathbb{P}_1$?

Linear Independence – Answers / Solutions

1. (a) Consider the vector equation

$$c_1 \sin t + c_2 \cos t = 0$$

for scalars $c_1, c_2 \in \mathbb{R}$. Note two functions are equal means they are equal for all values of t . Thus the equation must hold for particular t values. If $t = 0$, then we get the equation $c_1(0) + c_2(1) = 0$, and hence $c_2 = 0$. If $t = \pi/2$, then we get $c_1(1) + c_2(0) = 0$, and hence $c_1 = 0$. Thus the only solution to $c_1 \sin t + c_2 \cos t = 0$ is $c_1 = c_2 = 0$. Thus $\sin t$ and $\cos t$ are linearly independent.

- (b) Consider the vector equation

$$c_1(1 + t^2) + c_2(1 - t^2) = 0,$$

and regrouping terms we get

$$(c_1 + c_2) + t^2(c_1 - c_2) = 0.$$

Now equating coefficients we get $c_1 + c_2 = 0$ and $c_1 - c_2 = 0$, and hence $c_1 = c_2 = 0$. Thus $1 + t^2$ and $1 - t^2$ are linearly independent.

2. *Proof.* Let V be a vector space and let $S = \{v_1, \dots, v_p\} \subseteq V$. Let $H = \text{Span}\{v_1, \dots, v_p\}$.

(a) Without loss of generality, we assume v_p is a linear combination of $v_1, \dots, v_{p-1}$. Thus there exists $c_1, \dots, c_{p-1} \in \mathbb{R}$ such that

$$v_p = c_1 v_1 + \dots + c_{p-1} v_{p-1}.$$

Let $x \in \text{Span}\{v_1, \dots, v_p\}$, then there exists $d_1, \dots, d_p \in \mathbb{R}$ such that

$$x = d_1 v_1 + \dots + d_p v_p.$$

Plugging in our expression for v_p and using the vector space axioms we can simplify x in the following way:

$$x = d_1 v_1 + \dots + d_p v_p = d_1 v_1 + \dots + d_p (c_1 v_1 + \dots + c_{p-1} v_{p-1}) = (d_1 + d_p c_1) v_1 + \dots + (d_{p-1} + d_p c_{p-1}) v_{p-1}.$$

Thus we can express x as a linear combination of $v_1, \dots, v_{p-1}$, and hence $x \in \text{Span}\{v_1, \dots, v_{p-1}\}$.

(b) If S is linearly independent, then S is a basis for H , and hence the claim holds.

If S is linearly dependent, then some vector in S can be written as a linear combination of the others. By part (a), we can remove that vector from S without changing H . Repeat until we obtain a linearly independent set from S . Since H is non-zero, this process must end with some number of vectors from S . $\square$

3. *Proof.* Let U be the reduced row echelon form of A . The set of pivot columns of U is linearly independent since no column is a linear combination of the columns that precede it (by first Theorem today).

Since A is row equivalent to U , the pivot columns of A are also linearly independent.

Recall $\text{Col}(A) = \text{Span}\{a_1, \dots, a_n\}$. If a_k is a non-pivot column, then it is a linear combination of the other columns. By the spanning set theorem, we can remove each non-pivot column. Thus the pivot columns form a basis for the column space. $\square$

4. The first two columns are pivot columns, so by the previous theorem, we get

$$\text{Col}(A) = \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \right\}$$

and a basis is given by

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \right\}.$$

5. No, the vectors are not linearly independent. Note: $(1+t) + (1-t) = 2$ is a dependence relation. On the other hand, $(1+t)$ and $(1-t)$ do form a basis for $\mathbb{P}_1$. We would just need to check that they span $\mathbb{P}_1$ and they are linearly independent.